

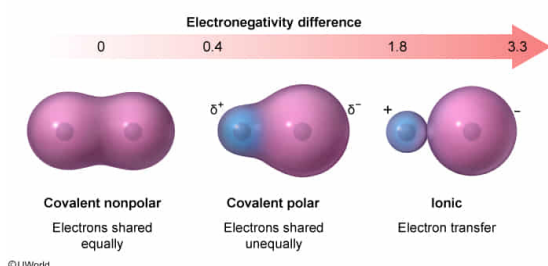
In chemical bond formation, covalent bonds are generally formed by electron:

- ✓ ☒ A. sharing between two atoms with a small or moderate electronegativity difference. [87%]
- ☐ B. sharing between two atoms with a large electronegativity difference. [9%]
- ☐ C. transfers between two atoms with a small or moderate electronegativity difference. [1%]
- ☐ D. transfers between two atoms with a large electronegativity difference. [1%]

Correct

88% answered correctly

Explanation:



Covalent bonds are formed between two atoms by sharing electrons through the end-to-end overlap of atomic orbitals. In contrast, **ionic bonds** are formed between two atoms when the valence electrons from one atom are transferred to another atom, resulting in charged ions that are held together by strong electrostatic attractions.

The type of bond that forms between two atoms depends on the **relative difference** in **electronegativity** between the atoms. A large difference in electronegativity promotes ionic bond formation, but a small difference promotes covalent bond formation. As a result, ionic bonds generally form between **a metal and a nonmetal**, whereas covalent bonds generally form between **two nonmetals**; however, there are some exceptions because the ionic character of a bond exists on a continuum.

Therefore, in chemical bond formation, covalent bonds are generally formed by electron sharing between two atoms with a small or moderate electronegativity difference.

(Choice B) Electron sharing cannot occur between two atoms with a large difference in electronegativity. This would cause the highly electronegative atom to take valence electrons (an electron transfer) from the atom with low electronegativity.

(Choices C and D) Chemical bonds formed by electron transfers are ionic, rather than covalent, bonds.

Educational objective:

The type of bond formed between two atoms depends on the relative difference in electronegativity between them. Atoms with a large difference in electronegativity (usually a metal and a nonmetal) form ionic bonds. Atoms with a small difference in electronegativity (usually two nonmetals) form covalent bonds.

Time spent:0 sec
Subject:
General Chemistry

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System:
Interactions of Chemical Substances